

ĐẠI HỌC QUỐC GIA THÀNH PHỐ HỒ CHÍ MINH
TRƯỜNG ĐẠI HỌC CÔNG NGHỆ THÔNG TIN
KHOA MẠNG MÁY TÍNH VÀ TRUYỀN THÔNG



Computer Networking

Lab 04: Crimping RJ45 Ethernet Cable

A. OVERVIEW

1.1 Objectives

- To understand various types of network cables and the procedures for crimping network cables according to the T-568A and T-568B standards.
- To crimp straight-through and crossover cables for data transmission between different types of devices.

1.2 Tools

- Network Cable (RJ45 cable).
- Dedicated Crimping Tool.
- Network Cable Tester.

1.3 Practical Environment

- Network cable, crimping tool, and testing equipment.
- Personal computer or laptop.

B. CABLE CRIMPING PRACTICAL EXERCISE

2.1 Basic Knowledge

Initially, there are two common network cable crimping standards: the T568A (Standard A) and T568B (Standard B) cable end standards. The figure below illustrates the color order for the two network cable crimping standards, A and B.

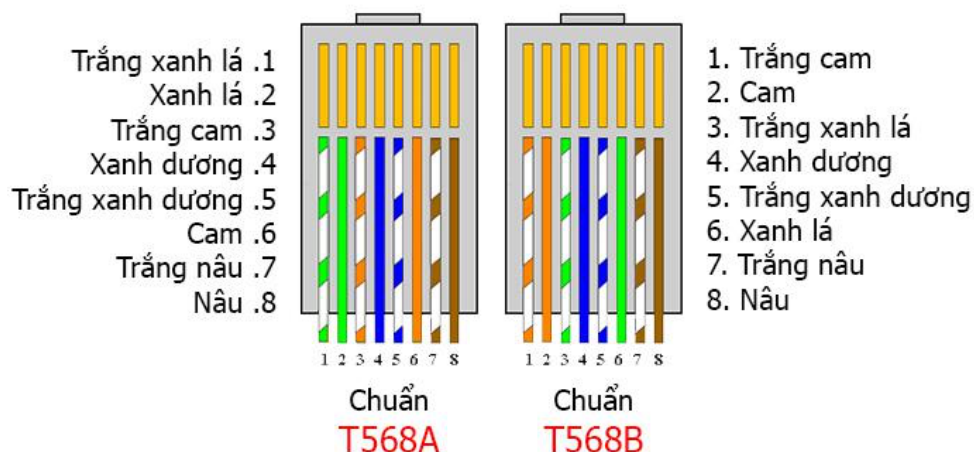


Fig 1. The Color Sequence of the Two Cable Crimping Standards

There are two styles of network cable crimping: straight-through cable crimping and crossover cable crimping. The appropriate crimping technique depends on the types of devices being connected.

- **Straight-Through Cable Crimping:** Both ends are crimped using the same standard (T568A or T568B). This type is used to connect two dissimilar devices.

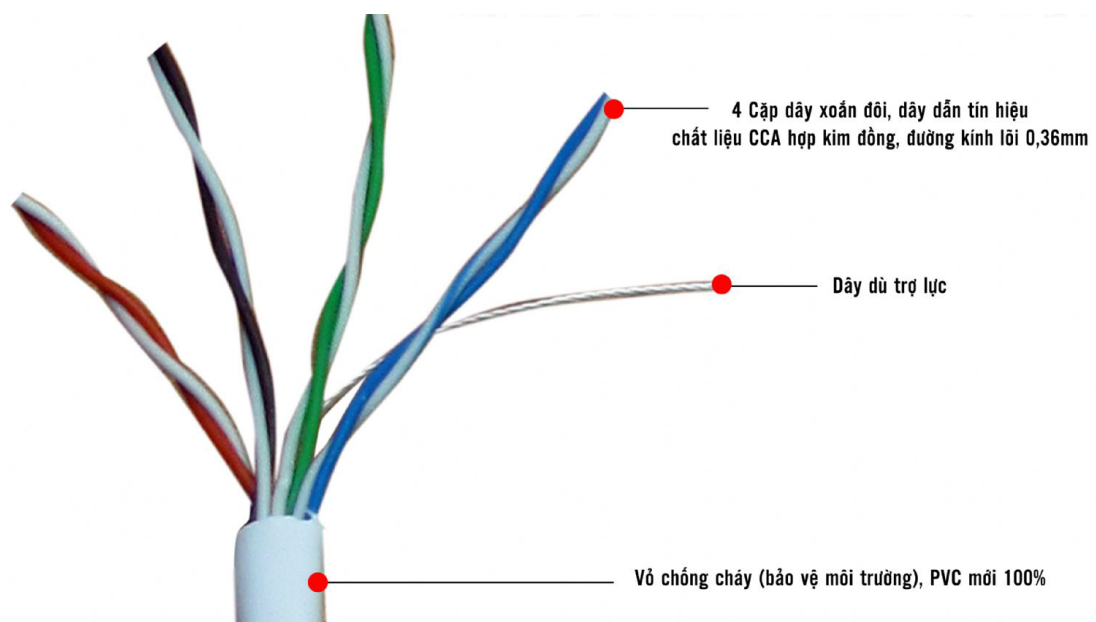
Examples: *PC + Switch, Switch + Router, etc*

- **Crossover Cable Crimping:** One cable end is crimped according to the T568A standard and the other end according to the T568B standard. This type is used to connect two similar devices.

Examples: *PC + PC, Switch + Switch, etc*

A standard compliant cable contains eight copper wires. These are twisted together into pairs: (Brown - White/Brown), (Orange - White/Orange), (Green - White/Green), (Blue - White/Blue). The cable also includes a metal/nylon thread.

Additionally, high-quality cables may have a sheath for protection against interference and heat.



The network cable is connected to an RJ45 connector. To crimp the cable, the eight copper wires must be inserted into the copper contacts within the RJ45 connector.

2.1 Network Cable Crimping Procedure

Since the straight-through crimping technique is relatively simple, students will practice crimping a crossover cable with one end using Standard A and the other end using Standard B:

- **Step 1:** Use the crimping tool or scissors to strip the cable jacket. Note: leave approximately 3-4 cm exposed to facilitate easier wire arrangement.

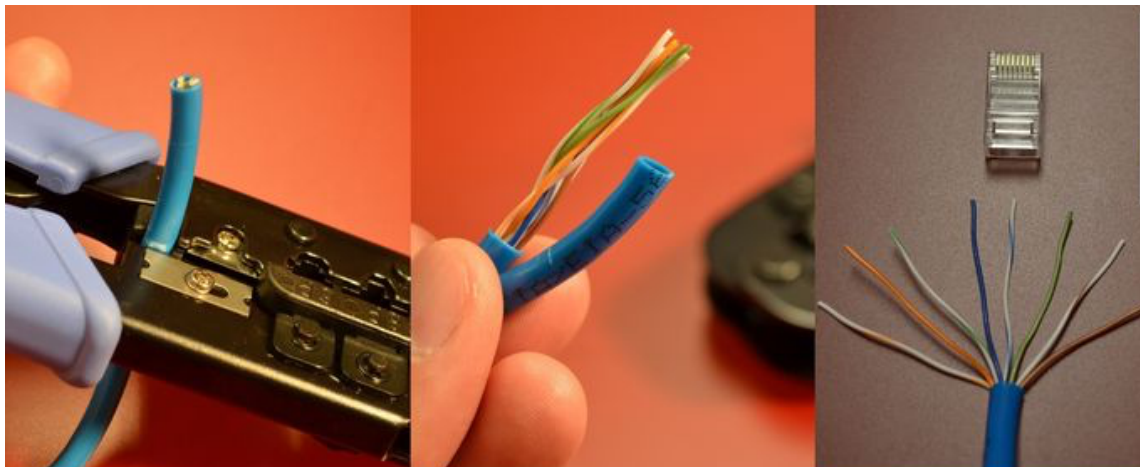


Fig 2. Utilize the recessed section of the crimping tool to rotate and strip a segment of the cable jacket at the end.

- **Step 2:** Untwist the pairs and arrange them in the following order:
White/Orange – Orange – White/Blue – Blue – White/Green – Green –
White/Brown – Brown.

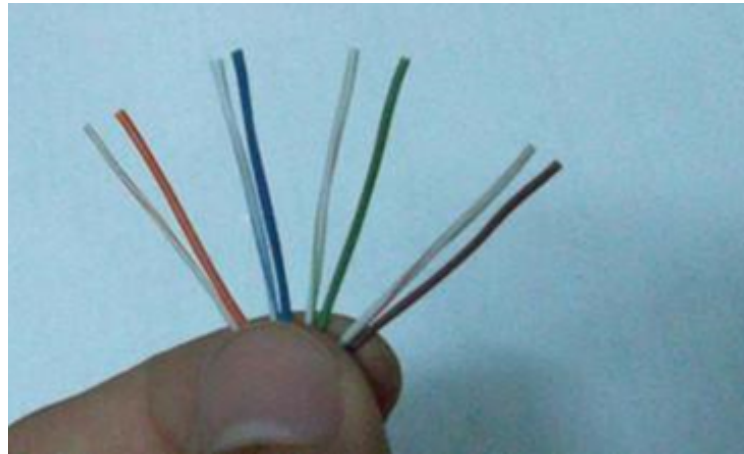


Fig 3. Arrange the cable wires in the sequence of White/Color followed by the corresponding Solid Color.

- **Step 3:** Swap the positions of the White/Green wire and the White/Blue wire.

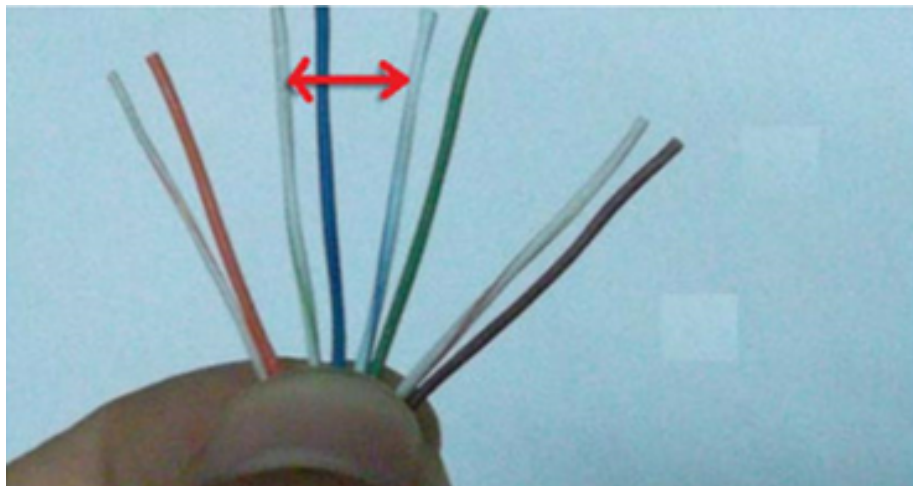


Fig 4. The resulting sequence for Standard B is: White/Orange – Orange – White/Green – Blue – White/Blue – Green – White/Brown – Brown.

- **Step 4:** Align the wires close together and use the crimping tool to cut them squarely and decisively. Ensure the cut is measured so that the portion of the cable with the jacket extends approximately 1/3 into the connector to securely fasten the connector and the cable.

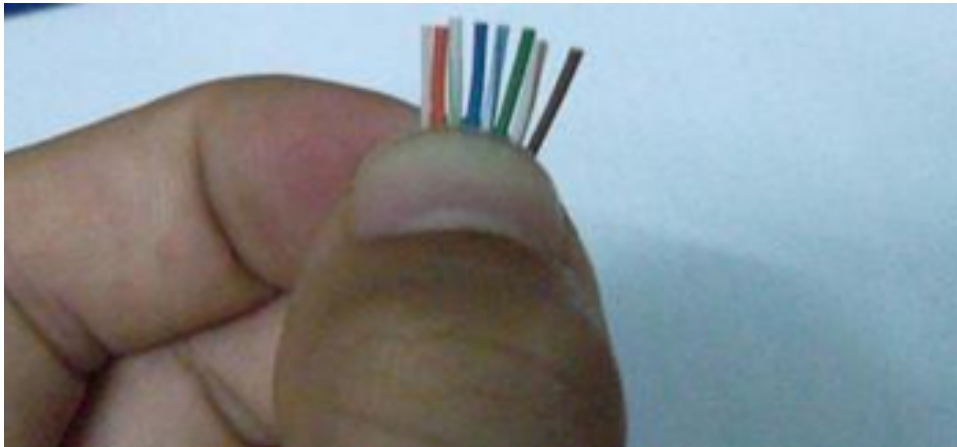


Fig 5. Ensure an even cut and maintain the correct sequence of the wire ends.

- **Step 5:** Insert the wires into the connector in the correct order, pushing them deeply so that the wire ends reach the copper core.

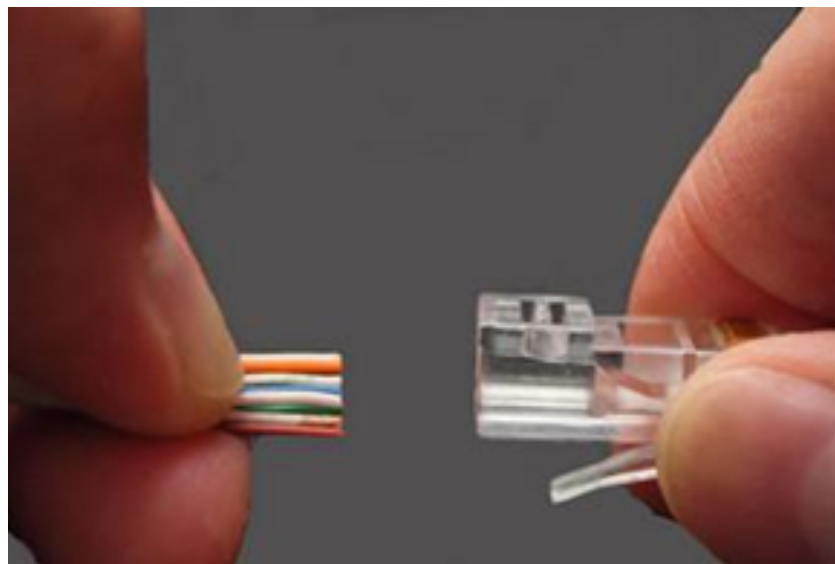


Fig 6. Ensure the correct order is maintained and the cable end is fully inserted until it reaches the copper core.

- **Step 6:** Place the connector into the crimping tool and firmly but moderately press the tool with a decisive motion.



Fig 7. Apply firm but moderate pressure during crimping; excessive force should be avoided.

- **Step 7:** Inspect to see if the cable's copper wires and the copper core inside the RJ45 connector are flush and secure.

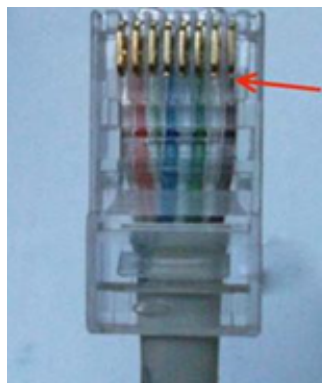


Fig 8. Image Illustrating a Successfully Crimped Cable End

- **Step 8:** Crimp the other cable end following the same steps, but for a crossover cable, the second end must use the opposite standard. If the first end was crimped to Standard B, the second end must be crimped to Standard A: White/Green – Green – White/Orange – Blue – White/Blue – Orange – White/Brown – Brown.
- **Step 9:** Test the crimped cable using a cable testing device.



Fig 9. Plug both cable ends into the two RJ45 ports of the device and turn it on.

For a correctly crimped straight-through cable, both parts of the tester will light up simultaneously in order from 1 to 8. For a correctly crimped crossover cable, both parts of the tester will light up in the following sequence: 1-3 → 2-6 → 3-1 → 4-4 → 5-5 → 6-2 → 7-7 → 8-8

If any number does not light up or does not follow the correct sequence, the crimping is considered unsuccessful.

2.2 Connecting Two Computers with an RJ45 Cable for Data Sharing

After successfully crimping and testing the cable, proceed to use the cable to connect two computers and share files between them. Assume two computers, A and B, using the Windows operating system.

- **Step 1:** Use the cable to connect the two computers via the RJ45 port.
- **Step 2:** Change the IP addresses of computers A and B to static IP addresses.

First, open the Network and Sharing Center.

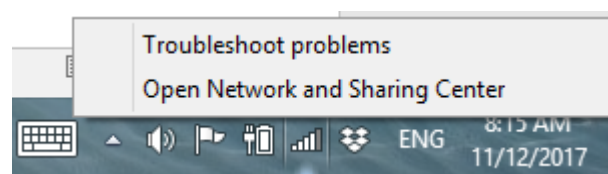


Fig 10. Right-click on the Network icon.

- **Step 3:** Select the active LAN connection > Ethernet > Select Properties.

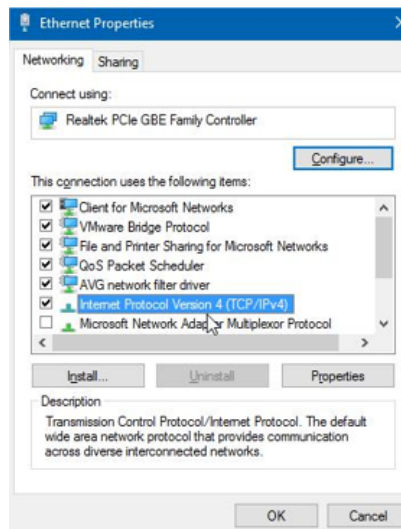


Fig 11. Double click on Internet Protocol Version 4 (TCP/IPv4)

- **Step 4:** Set the desired IP address.

Note: The IP must be within the Private IP range (IPs used exclusively for internal networks, inaccessible from the Internet), belonging to one of the following ranges:

10.0.0.0 - 10.255.255.255

172.16.0.0 - 172.31.255.255

192.168.0.0 - 192.168.255.255

For the two computers to connect, they must have two IPs within the same network class. Therefore, set both IPs to belong to the **192.168.1.x** network class (*each computer having a different value for x from 1-254*).

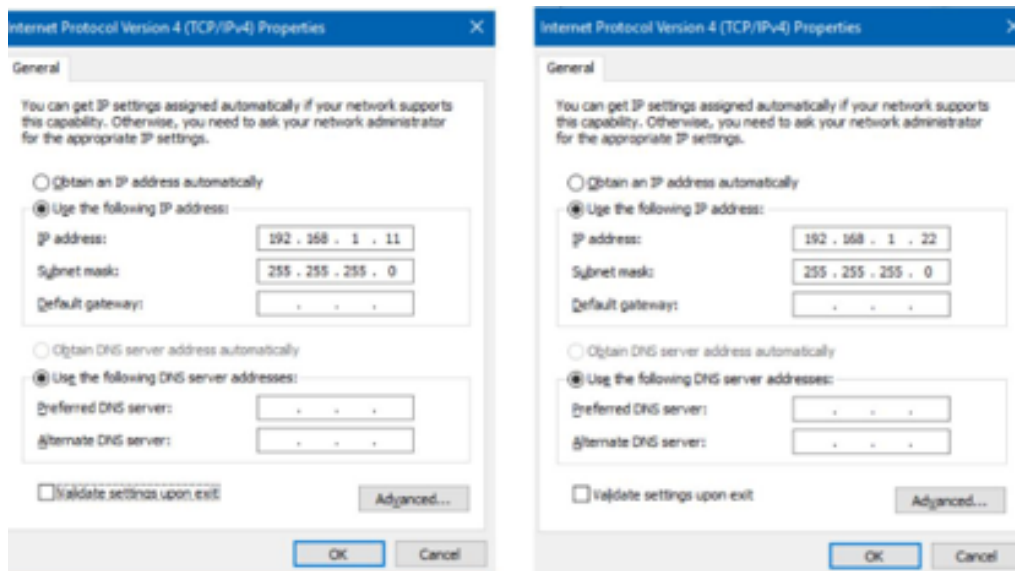


Fig 12. Example: Computer A set to IP = 192.168.1.11, Computer B set to IP = 192.168.1.22.

- **Step 5:** On the computer used for data sharing (e.g., computer A sharing data for computer B to access), it is recommended to disable the password-protected sharing option. Otherwise, when computer B connects to computer A's shared folder, it will need to enter computer A's login credentials. Disable password protection as follows:

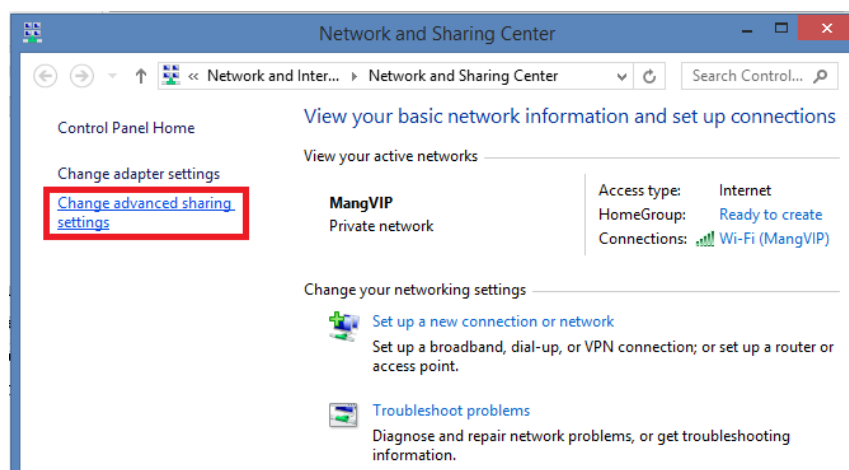


Fig 13. Select Change advanced sharing settings

- **Step 6:** Select All networks and turn off Password protected sharing.

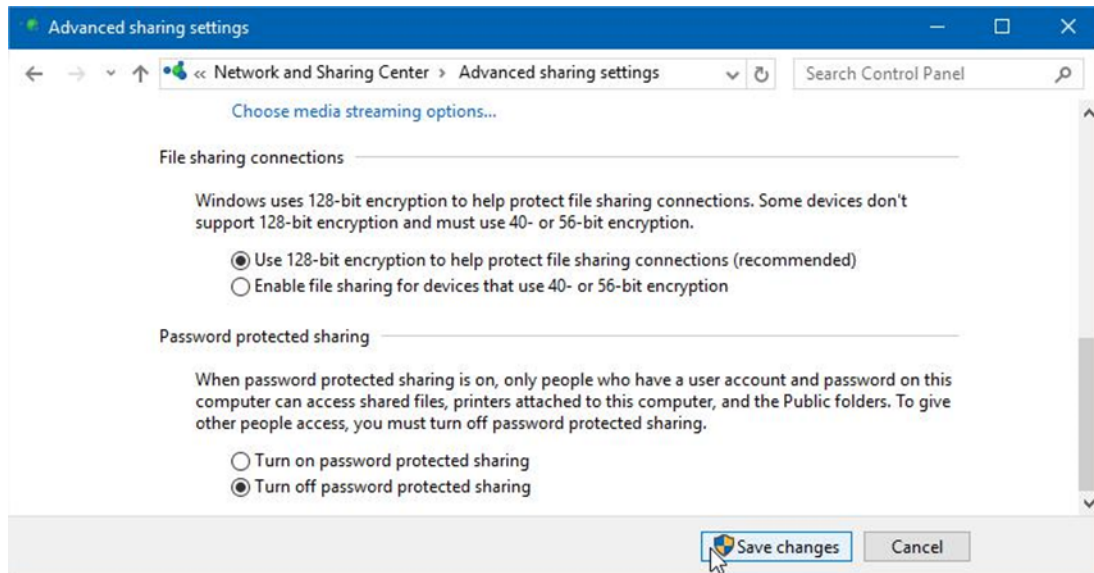


Fig 14. Disable the Password-Protected Sharing mode.

- **Step 7:** Select the folder to be shared, right-click, select Properties > Select Sharing > Advanced Sharing > Check Share this folder and select OK.

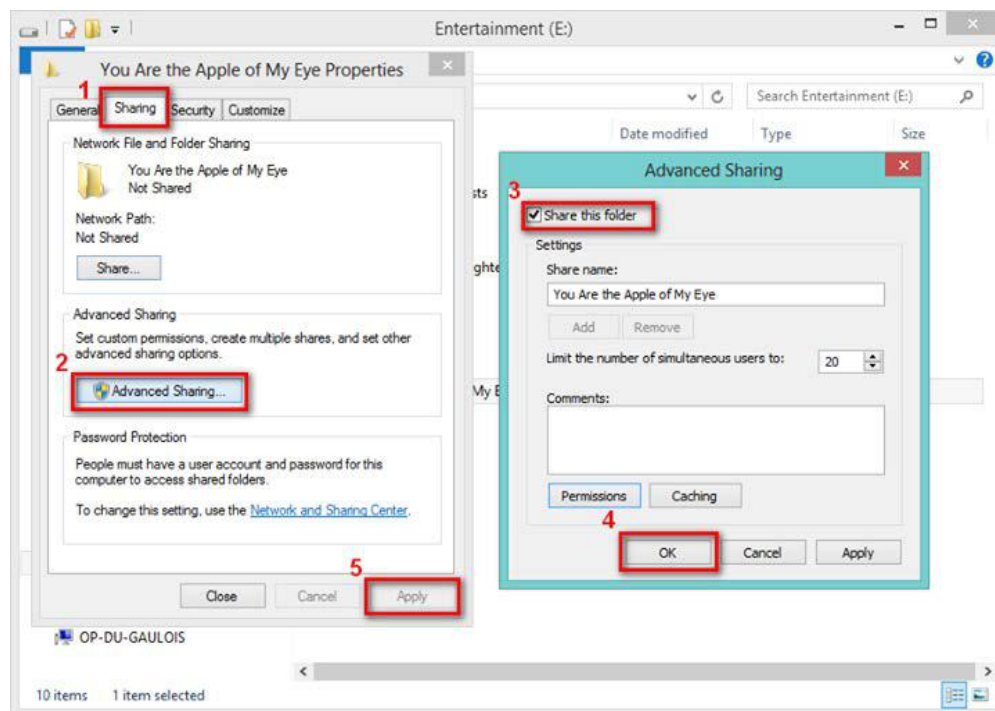


Fig 15. Folder Sharing from Host A

- **Step 8:** Computer B only needs to open Windows Explorer and access the path in the format \\192.168.1.11 (IP of computer A) to access the files that computer A has shared.

THE END
